WILSON PRANAJAYA TUNGGALA

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EDUCATION

Bina Nusantara University **Jakarta, Indonesia**
Bachelor of Science in Data Science ***2024 – Present***

- Current GPA: **3.81/4.00**

ORGANIZATIONAL EXPERIENCE

Data Science Club, Bina Nusantara University
Research and Development Activist ***2024 – Present***

- Conducted research on **Convolutional Neural Networks (CNN)** for multi-crop plant leaf disease classification using merged image datasets and deep learning experimentation.
- Co-authored the research paper *"CNN Architectures for Multi-Crop Leaf Disease Classification Using a Merged Dataset"*, published with IEEE, DOI: 10.1109/ITIS67966.2025.11308783.
- Explored Deep Learning algorithms and model architectures through Research & Development activities, strengthening practical experience in model experimentation, evaluation, and technical research.

CERTIFICATIONS

GCI World April 2026 Certificate of Completion **Aug 2026**
Matsuo-Iwasawa Laboratory, Graduate School of Engineering, The University of Tokyo

Fundamentals of Deep Learning **Mar 2025**
NVIDIA Credential ID: 1tO0Ys3ITkGJkXM3sgBKrQ

ACADEMIC AND INDEPENDENT PROJECTS

Real-Time Ride-Hailing Streaming Data Pipeline
Technologies: Python, PySpark, Apache Spark Structured Streaming, Redpanda, Redis, FastAPI, S3/LocalStack, H3, Streamlit, Docker

- Designed an end-to-end **Lambda Architecture** that streams NYC taxi events through Redpanda and Apache Spark while simultaneously serving real-time analytics through Redis and storing historical Parquet data in an S3-compatible data lake.
- Implemented event-time processing with **5-minute tumbling windows**, watermarking, and **H3 resolution 8** geospatial indexing to aggregate ride demand across geographic zones.
- Built a low-latency serving layer using Redis with a **10-minute TTL** and FastAPI, feeding a live 3D Streamlit/PyDeck dashboard that refreshes demand visualization every **3 seconds**.

Real-Time QRIS & E-Wallet Fraud Detection Pipeline
Technologies: Python, CatBoost, XGBoost, Redpanda, FastAPI, PostgreSQL, Streamlit, Docker

- Developed an event-driven fraud detection pipeline that streams PaySim transactions through Redpanda, performs real-time ML inference through FastAPI, persists prediction results in PostgreSQL, and exposes monitoring through a FraudOps dashboard.
- Compared XGBoost and CatBoost using time-series cross-validation; selected CatBoost after achieving a mean **PR-AUC of 0.7769**, compared with **0.6523** for XGBoost, representing approximately **19% higher PR-AUC**.
- Designed a business-oriented Top-K fraud investigation strategy that identified **9 of 10 fraudulent transactions** while requiring analysts to review only **150 of 50,000 transactions**, equivalent to reviewing just **0.3%** of the transaction volume.

AWS SageMaker Credit Score Prediction
Technologies: Python, Scikit-learn, XGBoost, LightGBM, CatBoost, AWS SageMaker, Amazon S3, EC2, Streamlit

- Developed an end-to-end credit scoring pipeline that classifies customers into **3 credit categories**: Poor, Standard, and Good, covering data ingestion, preprocessing, feature engineering, model training, evaluation, and cloud inference.
- Benchmarked **4 machine learning algorithms**: Random Forest, XGBoost, LightGBM, and CatBoost, then using stratified cross-validation and automated best-model selection based on weighted F1 score.
- Packaged the complete preprocessing and prediction pipeline for deployment to an **AWS SageMaker real-time endpoint**, integrated Amazon S3 model storage and an EC2-hosted Streamlit interface for interactive cloud inference.

SKILLS

Programming Languages: C (Basic), Python, SQL, R, JavaScript, HTML, CSS, PySpark

Data Engineering & Big Data: Apache Spark, Spark Structured Streaming, PySpark, Redpanda, Kafka-Compatible Streaming, ETL/ELT, Data Pipelines, Data Lakes, Parquet, Event Streaming, Batch Processing, Real-Time Processing, Data Preprocessing

Databases & Data Systems: SQL, PostgreSQL, Redis, Amazon S3

Machine Learning, Deep Learning & Statistics: Exploratory Data Analysis (EDA), Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, CatBoost, Convolutional Neural Networks (CNN), Image Classification, Computer Vision, NLP, Statistical Analysis, Descriptive Statistics, Hypothesis Testing, Feature Engineering

Data Visualization: Matplotlib, Seaborn, Streamlit, Microsoft Excel, Power BI

Cloud & MLOps: AWS SageMaker, Amazon S3, Amazon EC2, Docker, MLflow, FastAPI, Model Deployment, Real-Time Inference

Tools: Git, GitHub, Microsoft Office, Microsoft Excel

Soft Skills: Leadership, Analytical Thinking, Problem Solving, Communication, Teamwork, Adaptability, Initiative, Active Listening, Creative Thinking, Continuous Learning, Time Management, Collaboration, Presentation Skills

Languages: Indonesian (Native), English (Professional Working Proficiency)